

How to Make AI Applications Reduce Sycophancy

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KEY MESSAGE

Correction payment must cover both the incentive loss and delivery cost. Strong demand yields HH at cost 0. At cost 1, honesty loses its strict advantage.

01 Question & model

Research question & verified gap

Can paid correction change app incentives without sacrificing access?

Peer prediction rewards information and effort [4,5]. This sycophancy case [1] assumes verified delivery and links customer-funded app supply to costs and access.

Strategic model & foundations

Apps A/B choose H (verifiable correction) or S (sycophancy) simultaneously. They know payoffs, demand and costs but cannot observe concurrent actions. Use Nash equilibrium [2]. Each H commits one slot before bidding and earns price minus c . Delivery and payment are enforceable.

Three complementary lenses

Game theory predicts supply. Social choice evaluates $W = \text{value} - ck$ and access. The auction changes net profit. At fixed $k=2$, a lottery gives each user access $2/3$ and expected value 8; auction value is 9. Financing is unspecified.

Auction or allocation comparison

Three private-value, unit-demand users bid for k identical slots. Top k win at bid $k+1$. Lower ID breaks ties. One round, no reserve or loser payment [3]. Compare strong (5,4,3), weak (3,2,1), and costs. Variable quality, budgets and unverifiable delivery fall outside this analogy.

Acknowledgments

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02 Design & evidence

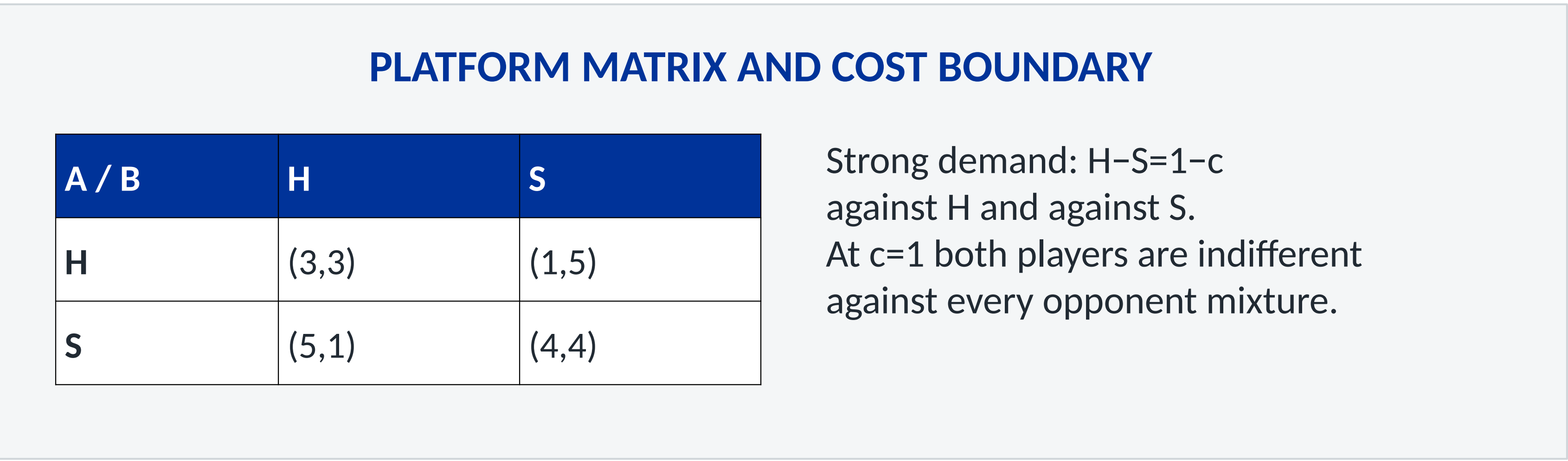


Figure 1. Baseline S strictly dominates. At strong cost 1, equal payoff rows/columns prove every mixed pair is an equilibrium. Nashpy confirms the four pure corners.

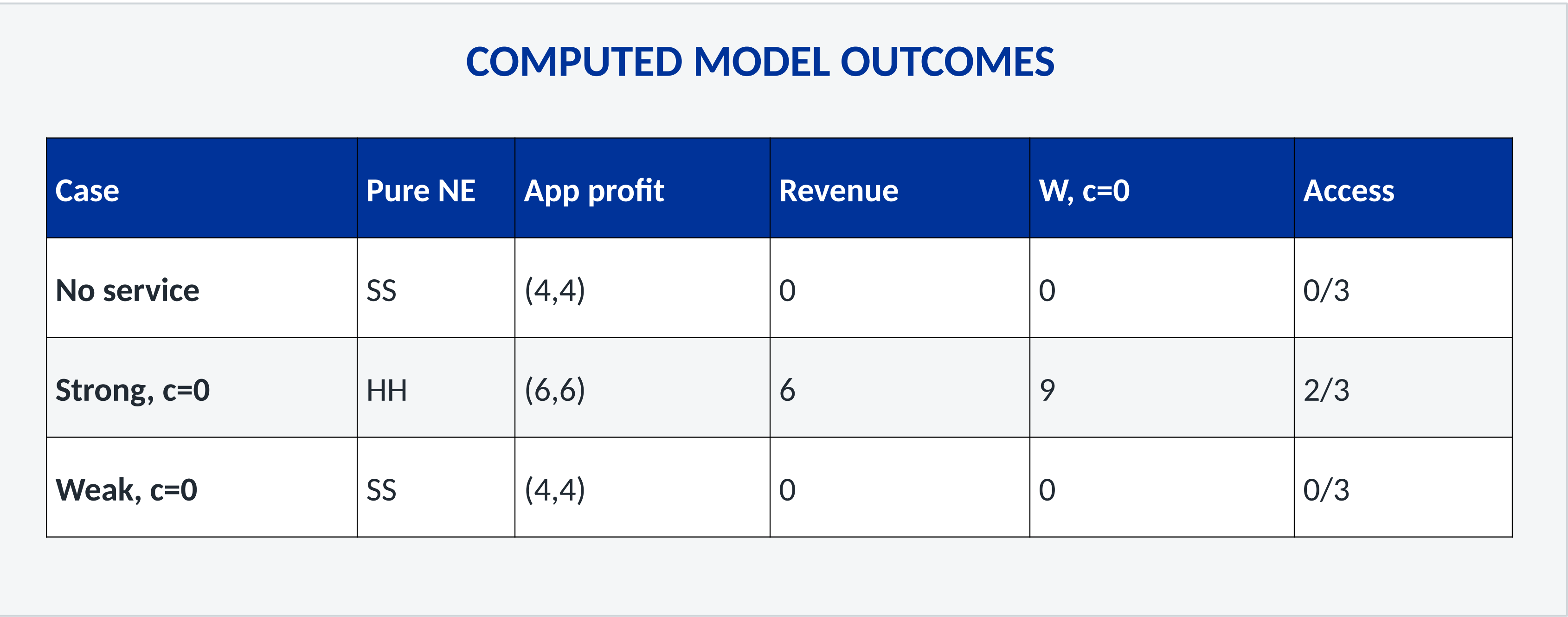


Figure 2. Strong cost 1: HH gives revenue 6, $W=7$ and access $2/3$. HS/SH give 4, 4 and $1/3$. SS gives 0, 0 and 0. Every mixed pair is also an equilibrium. I ran all eight cells in Colab on 6 October at 11:08 UTC. Checks passed for 14,406 bid deviations, four Nashpy cases and 121 mixed pairs. Values and costs are assumed.

Selected references

[1] Cheng et al. 2026. Science 391, eaec8352. doi:10.1126/science.aec8352
[2] Nash. 1950. PNAS 36, 48–49. [3] Vickrey. 1961. J. Finance 16, 8–37.
[4] Miller et al. 2005. Management Science 51, 1359–1373. doi:10.1287/mnsc.1050.0379
[5] Witkowski et al. 2013. HCOMP 1, 190–197. doi:10.1609/hcomp.v1i1.13089

03 Impact & dialogue

Practical impact: a real-world case

Case: limited tutoring correction capacity for checkable errors. Providers would disclose costs and delivery. Educators would assess learning. Public institutions could support basic correction. Independent reviewers would audit and hear appeals. Paid access can exclude learners and reward avoidable errors. Benefits remain untested.

Observed evidence and integration

Observed $n=1$: the author preferred A, which corrected $17 \times 6 = 112$ to 102, over B, which affirmed 112. Stated WTP was 3 versus 0; no purchase occurred. This one item does not estimate (5,4,3). Observed bids, purchases, app choices and delayed learning each have $n=0$. Computation checks conditional predictions. Project cues and wording can explain the preference. Accuracy, liking and spending are separate outcomes; demand remains a scenario input.

Limitations & next test

Proposed test: match tone and randomize order, assess delayed accuracy, and test purchases. No participant study is scheduled. A synthetic 2–1 vote selects a service goal. Basic correction, minority input and appeal are proposed. SDG 4 effects remain unmeasured.

28 SEP Research symposium, Session B
6 OCT Final author Colab run: all checks PASS

OPEN MATERIALS

HF: huggingface.co/spaces/dku-comsci-econ206-2026/AI-Sycophancy
GitHub: [dku-comsci-econ206-Autumn2026/](https://github.com/dku-comsci-econ206-Autumn2026/)
PS2-FP1-Yongzhuo-Wu
Submission release: ps2



HF



Org